

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

EE2002-1 – INSTRUMENTATION AND MEASUREMENTS
 (For EE)



Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Kelvin double bridge is modification of

A) Schering bridge	B) Maxwell's bridge
C) D'sauty's Bridge	D) Wheatstone bridge
2. A meter reads 127.5V and the true value of the voltage is 127.43 determine the static correction of the instrument

A) -0.7V	B) -0.08V
C) 0.09V	D) -0.07V
3. Precision is used in the measurements to describe the

A) consistency	B) redundancy
C) error	D) True value
4. 1mA meter movements with a internal resistance of 100 Ohm is to be converted into 0-100mA ammeter calculate the shunt required

A) 1 Ohm	B) 1.05 Ohm
C) 0.01 Ohm	D) 1.06 Ohm
5. Value of Multiplier should be invariant to

A) Voltage	B) temperature
C) current	D) resistance
6. In an energy meter, when pressure coil is energized and no current flows through the current coil and disc starts to rotate slowly. This condition is known as,

A) Control torque	B) Deflecting torque
C) creep	D) friction
7. Weston frequency meter has two coils _____ to each other

A) parallel	B) concentric
C) perpendicular	D) series
8. Phase sequence indicator works on the principle of _____

A) Self inductance	B) induction
C) DC motor	D) Flux leakage
9. Strain gauge should have _____ resistance temperature coefficient

A) moderate	B) low
C) high	D) zero
10. Transducer that converts linear motion to electrical signals

A) LVDT	B) Weston frequency meter
C) Strain Gauge	D) encoder

11. EMF obtained from a Hall effect transducer requires _____, before measure.
 A) amplification B) Inversion
 C) clipping D) Clamping
12. Bourdon tube senses _____
 A) temperature B) Linear displacement
 C) pressure D) friction
13. sensors that detect mechanical displacement for straight lines is
 A) Linear encoder B) Rotary encoders
 C) Strain guage D) LVDT
14. Thermistor is an _____ transducer
 A) Digital B) rotary
 C) linear D) analog
15. RTD have _____
 A) 100 Ohm B) zero
 C) infinite D) Positive temperature co-efficient
16. Thermistor have _____ sensitivity
 A) high B) moderate
 C) zero D) low
17. LCD are
 A) Dynamic scattering and active B) Dynamic scattering and passive
 C) Dynamic scattering and Field effect type D) Active and passive
18. LED is a PN junction device which emits light when current flows in _____ direction
 A) forward B) bidirection
 C) reverse D) Depletion region
19. P-N junction diode with appropriately doped semiconductors can be used as _____
 A) Strain gauge B) Photovoltaic cell
 C) LCD D) Polarizing layer
20. Lissajous pattern is used to measure
 A) Signal conditioning B) pressure
 C) frequency D) oscillation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

1. a) Write a short note on systematic error in an instrument.
 b) Explain with neat sketch the working principle of Kelvin Double Bridge.
2. a) It is desired to measure a value of current in the 500ohm resistor as shown in Figure 2a. determine the following (i) actual value of current (ii) measured value of current when an ammeter having resistance of 100 Ohm is connected in series with 500 Ohm (iii)percentage error (iv) Accuracy

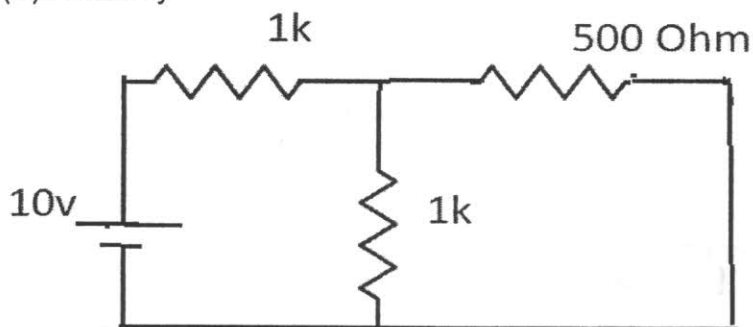


Fig 2(a)

Marks	BT*	CO*	PO*
6	L*2	1	1,2,10
4	L2	1	1,2,10
6	L2	2	1,2,10

6	L2	1	1,2,10
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b)	List the necessary qualities to be considered for designing a shunt to extend range in Ammeter.	4	L2	1	1,2,10
c)	Explain the operating principle of (i) moving system (ii) braking system (iii) Driving system in energy meter.	6	L3	2	1,2,10*
3. a)	Explain with a neat sketch the method used for localizing ground fault using Murray loop test.	4	L3	1	1,2,10
b)	With a neat sketch explain the working principle of Anderson's bridge.	6	L3	1	1,2,10
c)	Explain the purpose of using lag coil in energy meter.	6	L2	2	1,2,10
Unit – II					
4. a)	Write a short note on rotating type phase sequence indicator.	6	L2	3	1,2,10
b)	Explain the how LVDT and Bourdon tube can be used to measure pressure.	6	L2	3	1,2,10
c)	A thermistor has a resistance temperature coefficient of -5% over a temperature range of 25°C to 50°C. if the resistance is 100 Ohm at 25°C. what is the resistance at 35°C.	4	L2	4	1,2,10
5. a)	Mention the classification of strain gauge.	4	L2	3	1,2,10
b)	An output of an LVDT is connected to 5V voltmeter through an amplifier whose amplification factor is 250. An output of 2mV appears across the terminals of LVDT when the core moves through the distance of 0.5mm. Calculate the sensitivity of the LVDT and that of the the whole setup. The millivoltmeter scale has 100 divisions. The scale can be read to 1/5 of a division. Calculate the resolution of the instrument in mm.	6	L2	3	1,2,10
c)	Mention the disadvantages of the RTD.	6	L2	4	1,2,10
6. a)	Write a short note on gyroscope.	6	L2	4	1,2,10
b)	List the desirable characteristic of resistance wire strain Gauge.	4	L2	3	1,2,10
c)	Mention the difference between the magnetic and digital recording.	6	L2	4	1,2,10
Unit – III					
7. a)	With a neat sketch explain working of photovoltaic cell.	8	L3	5	1,2,10
b)	Mention the comparison between LCD and LED display.	8	L3	5	1,2,10
8. a)	With a neat block diagram explain working of digital storage oscilloscope.	8	L3	5	1,2,10
b)	List the difference between 7 segment and dot matrix display.	8	L3	5	1,2,10

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
